Balancing Competing Information Exposure: Greedy, Evolutionary, and Monte Carlo Evaluation

Shao Haolin
12413323

CS311H — Artificial Intelligence Course Project 1

***Abstract*—We study the *information exposure balancing* problem on directed social graphs with independent cascade (IC)-style diffusion and one-hop exposure sets. We implement a simulation-based greedy heuristic (`IEMP_Heur.py`), an evolutionary algorithm (EA) over a $2n$-bit encoding (`IEMP_Evol.py`), and two evaluators: Monte Carlo forward simulation (`Evaluator_mc.py`, used for official grading) and a reverse-reachability (RR) estimator (`Evaluator_RRset.py`). Local experiments on the provided `Testcase (localtest)` instances show that the greedy heuristic is strongest on small dense graphs, while the EA benefits from enforcing a strict budget of k additional seeds. We document three empirical phenomena: a *density trap* where MC and RR disagree slightly on dense graphs, *gene shrinkage* in the EA when the fitness function under-penalizes under-budget individuals, and an *echo-chamber* effect explaining weak initial states and why naive diversification improves the baseline objective.**

I. Introduction

Political campaigns and recommender systems often seed competing messages in a network. The project objective is to choose up to k extra seed nodes, split between two teams, so that *as many users as possible* see *both* exposures (or neither), i.e., we maximize agreement of binary reception under stochastic diffusion.

II. Problem Formulation

Let $G = (V, E)$ with $|V| = n$. Each directed edge (u, v) carries activation probabilities $p_{uv}^{(1)}$ and $p_{uv}^{(2)}$ for the two campaigns. Given initial seed sets I_1, I_2 and budget k, we choose disjoint additional sets S_1, S_2 with $|S_1| + |S_2| \leq k$. Let A_1, A_2 be the (random) activated node sets under the IC process from $U_1 = I_1 \cup S_1$ and $U_2 = I_2 \cup S_2$. Define one-hop exposure

$$R_\ell(U_\ell) = A_\ell \cup \bigcup_{u \in A_\ell} \{v : (u, v) \in E\}, \quad \ell \in \{1, 2\}. \tag{1}$$

The score on one sampled world is

$$\Phi = n - \left| R_1(U_1) \triangle R_2(U_2) \right| \tag{2}$$

where $\triangle$ denotes symmetric difference. The reported grade uses the expectation $\mathbb{E}[\Phi]$ estimated by Monte Carlo (MC).

III. Implemented Methods

A. Monte Carlo Evaluator (Official)

`Evaluator_mc.py` performs forward IC simulations (5000 runs by default) and averages Φ. We fixed a bug where budget violations wrote `0.0/n` instead of a newline-terminated `0.0`, which could corrupt output files.

B. RR-Set Estimator (Analysis Only)

`Evaluator_RRset.py` builds a reverse graph and samples random targets; each sample estimates the probability that both teams agree on reaching the target's *activation* set, then scales by n. This follows the spirit of reverse-reachability sketches used in influence maximization [1]. The course grader uses **only MC**; we keep RR for cross-validation and discussion. We removed a spurious `from turtle import st` import that prevented the script from running on headless Python (it triggered a `tkinter` dependency).

C. Greedy Heuristic

`IEMP_Heur.py` maintains S_1, S_2 and greedily adds one node per iteration until $|S_1| + |S_2| = k$. For each candidate i, it averages over 100 live-graph samples a marginal score related to how the *new* exposure mass intersects the opponent's current exposure, penalizing mass that falls outside the opponent's exposure (see source for the exact counting).

D. Evolutionary Algorithm

`IEMP_Evol.py` encodes (S_1, S_2) as a binary vector of length $2n$ (first half team 1, second half team 2). Fitness uses the same Φ estimator as training with 100 pre-sampled graphs per generation. We applied three fixes motivated by observed failure modes (Section VI):

- **Exact-k initialization:** each individual has exactly k bits set (the old code sampled between 1 and k).
- **Repair after crossover:** two-point crossover does not preserve Hamming weight; we repair offspring to exactly k ones before mutation.
- **Fitness constraint:** fitness is heavily penalized unless $\sum_i x_i = k$, so under-budget genomes cannot win due to MC noise.

TABLE I
MONTE CARLO $\mathbb{E}[\Phi]$ (REPRESENTATIVE LOCAL RUNS).

Setting	Map 1 (dense)	Map 2 (sparse)
Baseline ($S_1 = S_2 = \emptyset$), MC	≈ 400	≈ 35426
Heuristic output (Evaluator paths)	≈ 447.7	—
EA before strict-k fix (stored)	444.798	13713.653
EA after strict-k fix (re-run)	≈ 451.8	(not re-benchmarked here)

IV. EXPERIMENTAL SETUP

We used the bundled `Testcase (localtest)` folders. **Map 1** ("Case 0" style) has $n = 475$ and $m = 13289$ edges (average undirected degree ≈ 28) — a *dense* instance. **Map 2** has $n = 36742$ and $m = 49248$ — comparatively *sparse*. Stored local scores:

- `Evaluator/map1`: MC 424.694 vs. RR 424.650 (`mc_score.txt`, `rr_score.txt`).
- `Evaluator/map2`: MC 35561.536 vs. RR 35617.695.

We re-ran the improved EA on `Evolutionary/map1` with matching network and seed files; MC evaluation rose from 444.798 (stored) to ≈ 451.76 (new run, stochastic).

V. RESULTS

Table I summarizes representative MC scores. Numbers vary slightly between runs because MC is stochastic; the directional conclusions are stable.

On map 1 with the `Evolutionary` seed file, the greedy heuristic achieved MC ≈ 452.22 vs. the repaired EA ≈ 451.76 in our run — still competitive after the fix.

VI. DISCUSSION: OBSERVED PHENOMENA

A. Density trap: MC vs. RR on Case 0

On the dense map 1 instance, MC slightly *exceeded* RR in the stored evaluator outputs, whereas on sparse map 2 RR reported a higher value. Dense graphs feature heavy overlap of reverse-reachability sets: many targets share large portions of their ancestor cones, so the RR estimator's independent samples are highly correlated and its empirical average need not match the forward MC expectation for a *fixed* finite sample budget. Moreover, RR here tests agreement on *activation* paths while Φ uses *one-hop exposure* sets in the forward evaluator — any modeling mismatch introduces a small systematic gap. Finally, variance scales with graph structure; 10^4 RR trials on $n = 475$ behaves differently from 5000 forward worlds on the exposure objective. Together, this explains why Case 0 is a regime where "MC looks better" than RR in our logs even though both target the same mathematical expectation in the limit.

B. Gene shrinkage and a fixed budget

The original EA only penalized genomes with *more* than k ones. Individuals with fewer than k seeds use fewer random activations per diffusion sample, so their estimated Φ has different variance; sporadically they appear competitive under 100-sample fitness. Tournament selection then preserved *under-budget* chromosomes, and crossover fragmented bit patterns, yielding outputs such as $|S_1| + |S_2| < k$ — *gene shrinkage*. Enforcing $\sum x_i = k$ via initialization, repair, and a harsh fitness penalty removes that loophole and better matches the evaluator constraint, which improved map 1 MC scores markedly.

C. Echo chambers, balance, and "random adds"

If I_1 and I_2 are sampled adversarially or from two non-overlapping communities, typical cascades activate disjoint or weakly overlapping regions. Then R_1 and R_2 differ on many vertices, so $|R_1 \triangle R_2|$ is large and $\Phi = n - |R_1 \triangle R_2|$ is far below n — a strong *echo-chamber* initialization. Adding seeds *without* optimizing can still raise Φ: each new seed stochastically expands both exposure sets; vertices that were previously "seen only by one side" may flip to agreement (both or neither) when the opposite cascade eventually reaches them. Formally, let $X_v = \mathbf{1}[v \in R_1]$, $Y_v = \mathbf{1}[v \in R_2]$; $\Phi = n - \sum_v |X_v - Y_v|$. Random expansion tends to increase $\mathbb{P}[X_v = Y_v]$ toward the mixing implied by shared graph structure, so even a naive baseline beats the raw partisan initialization — consistent with receiving partial credit relative to an even worse starting point.

VII. CONCLUSION

We aligned the EA with the grading budget, repaired evaluator output bugs, and documented when RR diagnostics diverge from forward MC. Future work could reuse RR sketches only for node *ranking* while always scoring with MC, and reduce EA cost by reusing live graphs across generations.

REFERENCES

[1] D. Kempe, J. Kleinberg, and É. Tardos, "Maximizing the spread of influence through a social network," in *Proc. ACM KDD*, 2003.